

CSA0921 - Programming in Java

Level 3 Practice Programs

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1.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736`. The page header includes the SIMATS Engineering logo and the text "SIMATS Online Programming Lab" with "JAVA PROGRAMMING" below it. The user's name "Vidhushini T.S" is visible in the top right corner.

The main content area is titled "PRACTICE PROGRAMS-LEVEL 3 PROGRAMS". On the left, there is a "QUESTIONS" sidebar with a list of questions (Q1 to Q8). The selected question, Q1, is "Write a Java Program to Convert a Given Number of Days in Terms of Years, Weeks & Days." with a 10-mark value.

The question details include:

- Sample Input & Output:**
Enter the number of days: 756 No. of years: 2
No. of weeks: 3 No. of days: 5
- Test cases:**
1. 36
2. 36
3. 0
4. -365
5. -45

The "Your Code" section contains the following Java code:

```
1 import java.util.Scanner;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         int days = sc.nextInt();  
8  
9         int years = days / 365;  
10        days = days % 365;  
11  
12        int weeks = days / 7;  
13        int remainingDays = days % 7;  
14  
15        System.out.println("No. of years: " + years);  
16        System.out.println("No. of weeks: " + weeks);  
17        System.out.println("No. of days: " + remainingDays);  
18    }  
19 }
```

The "Input" field contains the value "756". The "Output" field displays the results:

```
No. of years: 2  
No. of weeks: 3  
No. of days: 5
```

At the bottom of the interface, there are "Run" and "Save" buttons. The Windows taskbar at the very bottom shows the time as 18:37 on 28-08-2026.

2.

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'CSA1607 SLOT D DWDM - JUL', 'SIMATS Online Lab', and 'Untitled document - Google D...'. The address bar shows 'simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736'. The page header features the SIMATS Engineering logo, the title 'SIMATS Online Programming Lab', and the user 'Vidhushini T S'. The main section is titled 'PRACTICE PROGRAMS-LEVEL 3 PROGRAMS'. On the left, a 'QUESTIONS' sidebar lists eight questions, with Q2 selected. Q2 asks to 'Develop a JAVA code to display the balance. Include the following members: Design a class to represent a bank account. Data Members: Name of the depositor, Account number, Type of account(Savings/Current), Balance amount in the account(Minimum balance is Rs.500.00). Methods: To read account number, Depositor name, Type of account. To deposit an amount (Deposited amount should be added with it). To withdraw an amount after checking balance(Minimum balance must be Rs.500.00 Note : Assume that balance amount = 10000). Testcase: 100, Raja, S, 8000; Raja, 100, S, 9000; 101, Rani, S, 12000; 102, Raghu, W, 8000; 103, Ravi, C, 10000'. The code editor on the right shows a partial implementation of the 'BankAccount' class with fields for account number, depositor name, account type, and balance. The 'Input' field contains '100 Raja S 8000' and the 'Output' field shows 'Balance: 8000.0'. The Windows taskbar at the bottom shows the date as 28-08-2026 and time as 18:37.

This screenshot shows the same SIMATS Online Programming Lab interface, but with the completed Java code in the editor. The code implements the 'BankAccount' class with methods for deposit, withdraw, and displayBalance. The 'main' method uses a Scanner to take input and create a BankAccount object. The 'Input' field now contains '100 Raja S 8000' and the 'Output' field shows 'Balance: 8000.0'. The 'QUESTIONS' sidebar on the left shows that 9 out of 9 questions have been answered. The footer of the page states '© 2026 SIMATS Online Lab. All rights reserved.' The Windows taskbar at the bottom shows the date as 28-08-2026 and time as 18:38.

3.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the user profile "Vidhushini T S".

The main content area is titled "PRACTICE PROGRAMS-LEVEL 3 PROGRAMS". On the left, there is a list of questions (Q1 to Q8) with their respective topics and marks. The selected question is Q3: "Develop a code to Reverse and Add a Number until you get a Palindrome." (10 marks).

The question details include:

- 3 Develop a code to Reverse and Add a Number until you get a Palindrome.
- Sample Input: If 7325 is input number, then
- 7325 (Input Number) + 5237 (Reverse Of Input Number) = 12562 12562 + 26521 = 39083
- 39083 + 38093 = 77176
- 77176 + 67177 = 144353
- 144353 + 353441 = 497794 (Palindrome)
- Test cases: 1. 8765, 2. -8765, 3. 0, 4. EIGHT FIVE, 5. 87.57

The "Your Code" section shows the following Java code:

```
import java.util.Scanner;

public class Main {

    static long reverse(long n) {
        long rev = 0;
        while (n > 0) {
            rev = rev * 10 + n % 10;
            n = n / 10;
        }
        return rev;
    }

    static boolean isPalindrome(long n) {
        return n == reverse(n);
    }
}
```

The "Input" field contains the value "7325", and the "Output" field displays "497794".

This screenshot shows the same SIMATS Online Programming Lab interface, but with the code editor expanded to show the full implementation, including the main method and the palindrome check logic.

The "Your Code" section now includes the following Java code:

```
import java.util.Scanner;

public class Main {

    static long reverse(long n) {
        long rev = 0;
        while (n > 0) {
            rev = rev * 10 + n % 10;
            n = n / 10;
        }
        return rev;
    }

    static boolean isPalindrome(long n) {
        return n == reverse(n);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        long n = sc.nextLong();
        if (n < 0) {
            n = Math.abs(n);
        }
        while (!isPalindrome(n)) {
            n = n + reverse(n);
        }
        System.out.println(n);
    }
}
```

The "Input" field still contains "7325", and the "Output" field still displays "497794". At the bottom of the code editor, there are "Run" and "Save" buttons.

4.

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'CSA1607 SLOT D DWDM - JUL', 'SIMATS Online Lab', and 'Untitled document - Google D...'. The address bar shows 'simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736'. The page header features the SIMATS Engineering logo, the title 'SIMATS Online Programming Lab', and a user profile 'Vidhushini T S'. The main content area is titled 'PRACTICE PROGRAMS-LEVEL 3 PROGRAMS'. On the left, a 'QUESTIONS' sidebar lists eight questions, with Q4 selected. Q4 asks to 'Write a program to print number of factors and to print nth factor of the given number.' with a 10-mark value. The main area displays the question details, sample input (100, N=4), sample output (9 factors, 4th factor is 5), and test cases. Below this, the 'Your Code' section shows a Java program that calculates the number of factors and the nth factor. The 'Input' field contains '100' and '4', and the 'Output' field shows 'Number of Factors = 9' and '4th factor of 100 = 5'. The bottom of the browser shows a Windows taskbar with a search bar and various application icons.

QUESTIONS

Q1 Write a Java Program to Convert a Given Number of Days in Terms of Years, Weeks & Days. 10 marks

Q2 Develop a Java code to display the balance. Include the following members: 10 marks

Q3 Develop a code to Reverse and Add a Number until you get a Palindrome. 10 marks

Q4 Write a program to print number of factors and to print nth factor of the given number. 10 marks

Q5 Reverse numbers 10 marks

Q6 Perfect squares 10 marks

Q7 Write a program to find the number of student users in the college, get the total users, staff user details from the client. Note for every 3 staff user there is one non-teaching staff user assigned by default. 10 marks

Q8 Given an integer num, return 10 marks

Q4 Write a program to print number of factors and to print nth factor of the given number. 10 marks

4 Write a program to print number of factors and to print nth factor of the given number.

Sample Input:
Given Number: 100
N = 4

Sample Output:
Number of factors = 9 4th factor of 100 = 5

Test cases:
1. Given Number = 512, N = 6
2. Given Number = 343, N = 7
3. Given Number = 1024, N = 0
4. Given Number = -6561, N = 3
5. Given Number = 0, N = 2

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int number = sc.nextInt();
8         int n = sc.nextInt();
9
10        if (number == 0) {
11            System.out.println("Number of factors = 0");
12            return;
13        }
14
15        int sum = Math.abs(number);
16        int count = 0;
17        int nthFactor = -1;
18
19        for (int i = 1; i <= sum; i++) {
20            if (sum % i == 0) {
21                count++;
22                if (count == n) {
23                    nthFactor = i;
24                }
25            }
26        }
27
28        System.out.println("Number of factors = " + count);
29        System.out.println(n + "th factor of " + number + " = " + nthFactor);
30    }
31 }
```

Input

100
4

Output

Number of Factors = 9
4th factor of 100 = 5

This screenshot is similar to the one above, showing the same SIMATS Online Programming Lab interface. The 'Your Code' section is expanded, showing the full Java program. The 'Input' and 'Output' fields are the same. The bottom of the browser shows a Windows taskbar with a search bar and various application icons.

QUESTIONS

Q4 Write a program to print number of factors and to print nth factor of the given number. 10 marks

Q5 Reverse numbers 10 marks

Q6 Perfect squares 10 marks

Q7 Write a program to find the number of student users in the college, get the total users, staff user details from the client. Note for every 3 staff user there is one non-teaching staff user assigned by default. 10 marks

Q8 Given an integer num, return 10 marks

Q9 Write a program to print unique permutations of a given number 10 marks

Q4 Write a program to print number of factors and to print nth factor of the given number. 10 marks

5. Given Number = 0, N = 2

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         int number = sc.nextInt();
8         int n = sc.nextInt();
9
10        if (number == 0) {
11            System.out.println("Number of factors = 0");
12            return;
13        }
14
15        int sum = Math.abs(number);
16        int count = 0;
17        int nthFactor = -1;
18
19        for (int i = 1; i <= sum; i++) {
20            if (sum % i == 0) {
21                count++;
22                if (count == n) {
23                    nthFactor = i;
24                }
25            }
26        }
27
28        System.out.println("Number of factors = " + count);
29        System.out.println(n + "th factor of " + number + " = " + nthFactor);
30    }
31 }
```

Input

100
4

Output

Number of Factors = 9
4th factor of 100 = 5

Run Save

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5.

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'SIMATS Online Lab' and 'Untitled document - Google D...'. The URL is 'simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736'. The page header features the SIMATS Engineering logo, the title 'SIMATS Online Programming Lab', and a user profile 'Vidhushini T S'. The main content area is titled 'PRACTICE PROGRAMS-LEVEL 3 PROGRAMS' and contains a list of questions on the left and a detailed question on the right.

QUESTIONS

- Q1: Write a Java Program to Convert a Given Number of Days in Terms of Weeks & Days. (10 marks)
- Q2: Develop a JAW code to display the balance. Include the following members: (10 marks)
- Q3: Develop a code to Reverse and Add a Number until you get a Palindrome. (10 marks)
- Q4: Write a program to print number of factors and to print each factor of the given number. (10 marks)
- Q5: Roman numerals (10 marks)
- Q6: perfect squares (10 marks)
- Q7: Write a program to find the number of student users in the college, get the total users, staff user details from the client. Note for every staff user there is one non teaching staff user assigned by default. (10 marks)
- Q8: Given an integer num, return (10 marks)

Q5: Roman numerals (10 marks)

5 Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.

Symbol Value

I	1
V	5
X	10
L	50
C	100
D	500
M	1000

For example, 2 is written as II in Roman numeral, just two ones added together. 12 is written as XII, which is X + II. The number 27 is written as XXVII, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:

- I can be placed before V (5) and X (10) to make 4 and 9.
- X can be placed before L (50) and C (100) to make 40 and 90.
- C can be placed before D (500) and M (1000) to make 400 and 900.

Given a roman numeral, convert it to an integer. Example:

Input: s = "III" Output: 3

Test cases:

1. LVIII
2. MCMXCIV
3. V
4. LXXII
5. MCCCLIV

Your Code

```
import java.util.Scanner;

public class Main {

    static int value(char c) {

        switch (c) {
```

Input

III

Output

The screenshot shows the same SIMATS Online Programming Lab interface, but with the code for the 'Roman numerals' question completed. The 'Your Code' section now contains a full Java program that converts a Roman numeral to an integer.

Q8: Given an integer num, return (10 marks)

is one non teaching staff user assigned by default. (10 marks)

Q9: Write a program to print unique permutations of a given number (10 marks)

9/9 answered

Q5: Roman numerals (10 marks)

5 Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.

Symbol Value

I	1
V	5
X	10
L	50
C	100
D	500
M	1000

For example, 2 is written as II in Roman numeral, just two ones added together. 12 is written as XII, which is X + II. The number 27 is written as XXVII, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:

- I can be placed before V (5) and X (10) to make 4 and 9.
- X can be placed before L (50) and C (100) to make 40 and 90.
- C can be placed before D (500) and M (1000) to make 400 and 900.

Given a roman numeral, convert it to an integer. Example:

Input: s = "III" Output: 3

Test cases:

1. LVIII
2. MCMXCIV
3. V
4. LXXII
5. MCCCLIV

Your Code

```
import java.util.Scanner;

public class Main {

    static int value(char c) {

        switch (c) {
            case 'I': return 1;
            case 'V': return 5;
            case 'X': return 10;
            case 'L': return 50;
            case 'C': return 100;
            case 'D': return 500;
            case 'M': return 1000;
            default: return 0;
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String roman = sc.nextLine().trim().toUpperCase();

        int result = 0;

        for (int i = 0; i < roman.length(); i++) {
            int current = value(roman.charAt(i));

            if (i + 1 < roman.length()
                && current < value(roman.charAt(i + 1))) {
                result -= current;
            } else {
                result += current;
            }
        }

        System.out.println(result);
    }
}
```

Input

III

Output

3

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6.

The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the page title "SIMATS Online Programming Lab", and a user profile "Vidhushini T S". The main content area is titled "PRACTICE PROGRAMS-LEVEL 3 PROGRAMS". On the left, a "QUESTIONS" sidebar lists eight questions, with "Q6 perfect squares" selected. The question text is: "6 Write a Program to create a list of all numbers in a range which are perfect squares and the sum of the digits of the number is less than 10. Sample input & Output: Enter lower range: 40 [1, 4, 9, 16, 25, 36]". Below the question, the "Your Code" section contains a Java program. The "Input" field shows "1" and "40", and the "Output" field shows "11, 4, 9, 16, 25, 36".

QUESTIONS

- Q1 Write a Java Program to Convert a Given Number of Days in Terms of Weeks, Months & Days. 10 marks
- Q2 Develop a Java code to display the balance, include the following members: 10 marks
- Q3 Develop a code to Reverse and Add a Number until you get a Palindrome. 10 marks
- Q4 Write a program to print number of factors and to print each factor of the given number. 10 marks
- Q5 Reverse numbers. 10 marks
- Q6 perfect squares. 10 marks
- Q7 Write a program to find the number of student users in the college, get the total users, staff user details from the client. Note for every 3 staff user there is one non-teaching staff user assigned by default. 10 marks
- Q8 Given an integer num, return the number of digits to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it. 10 marks

Q6 perfect squares 10 marks

6 Write a Program to create a list of all numbers in a range which are perfect squares and the sum of the digits of the number is less than 10.
Sample input & Output: Enter lower range: 40 [1, 4, 9, 16, 25, 36]

Test cases:
1. Enter lower range: 50 Enter upper range: 100
2. Enter lower range: 5 Enter upper range: 8
3. Enter lower range: 10 Enter upper range: 5
4. Enter lower range: 500 Enter upper range: 500
5. Enter lower range: 0 Enter upper range: -100

Your Code

```

1 import java.util.Scanner;
2
3 public class Main {
4
5     static int digitSum(int n) {
6         int sum = 0;
7         while (n > 0) {
8             sum += n % 10;
9             n /= 10;
10        }
11        return sum;
12    }
13
14    public static void main(String[] args) {
15        Scanner sc = new Scanner(System.in);
16
17        int lower = sc.nextInt();
18        int upper = sc.nextInt();
19
20        if (lower > upper) {
21            System.out.println("!");
22            return;
23        }
24    }
25
26    public static void main(String[] args) {
27        Scanner sc = new Scanner(System.in);
28
29        int lower = sc.nextInt();
30        int upper = sc.nextInt();
31
32        if (lower > upper) {
33            System.out.println("!");
34            return;
35        }
36    }
37
38    public static void main(String[] args) {
39        Scanner sc = new Scanner(System.in);
40
41        int lower = sc.nextInt();
42        int upper = sc.nextInt();
43
44        if (lower > upper) {
45            System.out.println("!");
46            return;
47        }
48    }
49
50    public static void main(String[] args) {
51        Scanner sc = new Scanner(System.in);
52
53        int lower = sc.nextInt();
54        int upper = sc.nextInt();
55
56        if (lower > upper) {
57            System.out.println("!");
58            return;
59        }
60    }
61
62    public static void main(String[] args) {
63        Scanner sc = new Scanner(System.in);
64
65        int lower = sc.nextInt();
66        int upper = sc.nextInt();
67
68        if (lower > upper) {
69            System.out.println("!");
70            return;
71        }
72    }
73
74    public static void main(String[] args) {
75        Scanner sc = new Scanner(System.in);
76
77        int lower = sc.nextInt();
78        int upper = sc.nextInt();
79
80        if (lower > upper) {
81            System.out.println("!");
82            return;
83        }
84    }
85
86    public static void main(String[] args) {
87        Scanner sc = new Scanner(System.in);
88
89        int lower = sc.nextInt();
90        int upper = sc.nextInt();
91
92        if (lower > upper) {
93            System.out.println("!");
94            return;
95        }
96    }
97
98    public static void main(String[] args) {
99        Scanner sc = new Scanner(System.in);
100       int lower = sc.nextInt();
101       int upper = sc.nextInt();
102
103       if (lower > upper) {
104           System.out.println("!");
105           return;
106       }
107   }
108
109   public static void main(String[] args) {
110       Scanner sc = new Scanner(System.in);
111
112       int lower = sc.nextInt();
113       int upper = sc.nextInt();
114
115       if (lower > upper) {
116           System.out.println("!");
117           return;
118       }
119   }
120
121   public static void main(String[] args) {
122       Scanner sc = new Scanner(System.in);
123
124       int lower = sc.nextInt();
125       int upper = sc.nextInt();
126
127       if (lower > upper) {
128           System.out.println("!");
129           return;
130       }
131   }
132
133   public static void main(String[] args) {
134       Scanner sc = new Scanner(System.in);
135
136       int lower = sc.nextInt();
137       int upper = sc.nextInt();
138
139       if (lower > upper) {
140           System.out.println("!");
141           return;
142       }
143   }
144
145   public static void main(String[] args) {
146       Scanner sc = new Scanner(System.in);
147
148       int lower = sc.nextInt();
149       int upper = sc.nextInt();
150
151       if (lower > upper) {
152           System.out.println("!");
153           return;
154       }
155   }
156
157   public static void main(String[] args) {
158       Scanner sc = new Scanner(System.in);
159
160       int lower = sc.nextInt();
161       int upper = sc.nextInt();
162
163       if (lower > upper) {
164           System.out.println("!");
165           return;
166       }
167   }
168
169   public static void main(String[] args) {
170       Scanner sc = new Scanner(System.in);
171
172       int lower = sc.nextInt();
173       int upper = sc.nextInt();
174
175       if (lower > upper) {
176           System.out.println("!");
177           return;
178       }
179   }
180
181   public static void main(String[] args) {
182       Scanner sc = new Scanner(System.in);
183
184       int lower = sc.nextInt();
185       int upper = sc.nextInt();
186
187       if (lower > upper) {
188           System.out.println("!");
189           return;
190       }
191   }
192
193   public static void main(String[] args) {
194       Scanner sc = new Scanner(System.in);
195
196       int lower = sc.nextInt();
197       int upper = sc.nextInt();
198
199       if (lower > upper) {
200           System.out.println("!");
201           return;
202       }
203   }
204
205   public static void main(String[] args) {
206       Scanner sc = new Scanner(System.in);
207
208       int lower = sc.nextInt();
209       int upper = sc.nextInt();
210
211       if (lower > upper) {
212           System.out.println("!");
213           return;
214       }
215   }
216
217   public static void main(String[] args) {
218       Scanner sc = new Scanner(System.in);
219
220       int lower = sc.nextInt();
221       int upper = sc.nextInt();
222
223       if (lower > upper) {
224           System.out.println("!");
225           return;
226       }
227   }
228
229   public static void main(String[] args) {
230       Scanner sc = new Scanner(System.in);
231
232       int lower = sc.nextInt();
233       int upper = sc.nextInt();
234
235       if (lower > upper) {
236           System.out.println("!");
237           return;
238       }
239   }
240
241   public static void main(String[] args) {
242       Scanner sc = new Scanner(System.in);
243
244       int lower = sc.nextInt();
245       int upper = sc.nextInt();
246
247       if (lower > upper) {
248           System.out.println("!");
249           return;
250       }
251   }
252
253   public static void main(String[] args) {
254       Scanner sc = new Scanner(System.in);
255
256       int lower = sc.nextInt();
257       int upper = sc.nextInt();
258
259       if (lower > upper) {
260           System.out.println("!");
261           return;
262       }
263   }
264
265   public static void main(String[] args) {
266       Scanner sc = new Scanner(System.in);
267
268       int lower = sc.nextInt();
269       int upper = sc.nextInt();
270
271       if (lower > upper) {
272           System.out.println("!");
273           return;
274       }
275   }
276
277   public static void main(String[] args) {
278       Scanner sc = new Scanner(System.in);
279
280       int lower = sc.nextInt();
281       int upper = sc.nextInt();
282
283       if (lower > upper) {
284           System.out.println("!");
285           return;
286       }
287   }
288
289   public static void main(String[] args) {
290       Scanner sc = new Scanner(System.in);
291
292       int lower = sc.nextInt();
293       int upper = sc.nextInt();
294
295       if (lower > upper) {
296           System.out.println("!");
297           return;
298       }
299   }
300
301   public static void main(String[] args) {
302       Scanner sc = new Scanner(System.in);
303
304       int lower = sc.nextInt();
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1238      Scanner sc = new Scanner(System.in);
1239
1240      int lower = sc.nextInt();
1241      int upper = sc.nextInt();
1242
1243      if (lower > upper) {
1244          System.out.println("!");
1245          return;
1246      }
1247  }
1248
1249  public static void main(String[] args) {
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1281          return;
1282      }
1283  }
1284
1285  public static void main(String[] args) {
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1288      int lower = sc.nextInt();
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1290
1291      if (lower > upper) {
1292          System.out.println("!");
1293          return;
1294      }
1295  }
1296
1297  public static void main(String[] args) {
1298      Scanner sc = new Scanner(System.in);
1299
1300      int lower = sc.nextInt();
1301      int upper = sc.nextInt();
1302
1303      if (lower > upper) {
1304          System.out.println("!");
1305          return;
1306      }
1307  }
1308
1309  public static void main(String[] args) {
1310      Scanner sc = new Scanner(System.in);
1311

```

7.

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'CSA1607 SLOT D DWDM - JUL', 'SIMATS Online Lab', and 'Untitled document - Google D...'. The address bar shows 'simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736'. The page header features the SIMATS Engineering logo, the title 'SIMATS Online Programming Lab', and a user profile 'Vidhushini T S'. The main section is titled 'PRACTICE PROGRAMS-LEVEL 3 PROGRAMS'. On the left, a 'QUESTIONS' sidebar lists eight questions, with Question 7 selected. Question 7 asks: 'Write a program to find the number of student users in the college, get the total users, staff users details from the client. Note for every 3 staff user there is one Non teaching staff user assigned by default.' The question is worth 10 marks. The problem description includes sample input (Total Users: 856, Staff Users: 126) and sample output (Student Users: 688). Test cases are provided: 1. Total User: 0, 2. Total User: -143, 3. Total User: 1026, Staff User: 1026, 4. Total User: 450, Staff User: 540, 5. Total User: 600, Staff User: 450. The 'Your Code' section contains a Java program that calculates the number of student users based on the total users and staff users, applying the 3:1 ratio for non-teaching staff. The 'Input' field shows '856' and '126', and the 'Output' field shows 'Student Users: 688'.

8.

The screenshot shows the SIMATS Online Programming Lab interface for Question 8. The browser tabs and address bar are the same as in the previous screenshot. The main section is titled 'PRACTICE PROGRAMS LEVEL 3 PROGRAMS'. On the left, the 'QUESTIONS' sidebar lists eight questions, with Question 8 selected. Question 8 asks: 'Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.' The question is worth 10 marks. The problem description includes an example: Input: num = 14, Output: 6. The steps are: Step 1) 14 is even, divide by 2 and obtain 7. Step 2) 7 is odd, subtract 1 and obtain 6. Step 3) 6 is even, divide by 2 and obtain 3. Step 4) 3 is odd, subtract 1 and obtain 2. Step 5) 2 is even, divide by 2 and obtain 1. Step 6) 1 is odd, subtract 1 and obtain 0. Test cases are provided: 1. n = 5, 2. n = 10, 3. n = 12, 4. n = 18, 5. n = 20. The 'Your Code' section contains a Java program that implements the step-counting logic using a while loop. The 'Input' field shows '14', and the 'Output' field shows '6'.

9.

The screenshot shows the SIMATS Online Programming Lab interface. The browser tabs include 'CSA1607 SLOT D DWDM - JUL', 'SIMATS Online Lab', and 'Untitled document - Google D...'. The address bar shows 'simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1736'. The page header features the SIMATS Engineering logo, the title 'SIMATS Online Programming Lab', and a user profile 'Vidhushini T S'. The main section is titled 'PRACTICE PROGRAMS-LEVEL 3 PROGRAMS'. On the left, a 'QUESTIONS' sidebar lists eight questions (Q1-Q8) with their respective marks. The main content area displays question Q9: 'Write a program to print unique permutations of a given number' (10 marks). It provides sample input (143) and sample output (134, 143, 314, 341, 413, 431). Below this, there are fields for 'Your Code' and 'Input'. The 'Your Code' field contains a Java program for generating permutations. The 'Input' field contains the number 143. The 'Output' field is empty.

QUESTIONS

- Q1 Write a Java Program to Convert a Given Number of Days in Terms of Years, Weeks & Days. 10 marks
- Q2 Develop a JAWA code to display the balance. Include the following members: 10 marks
- Q3 Develop a code to Reverse and Add a Number until you get a Palindrome. 10 marks
- Q4 Write a program to print number of factors and to print each factor of the given number. 10 marks
- Q5 Repeat numbers 10 marks
- Q6 perfect squares 10 marks
- Q7 Write a program to find the number of student users in the college, get the total users, staff user details from the client. Note for every 3 staff user there is one from teaching staff user assigned by default. 10 marks
- Q8 Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it. 10 marks

Q9 Write a program to print unique permutations of a given number 10 marks

Sample Input:
Given Number: 143

Sample Output:
Permutations are:
134
143
314
341
413
431

Test cases:
1. 0
2. 111
3. 505
4. -143
5. -598

Your Code

```
import java.util.Scanner;

public class Main {

    static void permute(char[] arr, int left, int right) {

        if (left == right) {
            System.out.println(new String(arr));
            return;
        }

        boolean[] used = new boolean[10];

        for (int i = left; i <= right; i++) {

            import java.util.Scanner;

            public class Main {

                static void permute(char[] arr, int left, int right) {

                    if (left == right) {
                        System.out.println(new String(arr));
                        return;
                    }

                    boolean[] used = new boolean[10];

                    for (int i = left; i <= right; i++) {
```

Input
143

Output

This screenshot shows the same SIMATS Online Programming Lab interface, but with the 'Your Code' field filled with the complete Java program for generating permutations. The 'Output' field now displays the results of the program execution for the input 143.

QUESTIONS

- Q7 Write a program to find the number of student users in the college, get the total users, staff user details from the client. Note for every 3 staff user there is one from teaching staff user assigned by default. 10 marks
- Q8 Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it. 10 marks
- Q9 Write a program to print unique permutations of a given number. 10 marks

9/9 answered

Q9 Write a program to print unique permutations of a given number 10 marks

Sample Input:
Given Number: 143

Sample Output:
Permutations are:
134
143
314
341
413
431

Your Code

```
static void permute(char[] arr, int left, int right) {

    if (left == right) {
        System.out.println(new String(arr));
        return;
    }

    boolean[] used = new boolean[10];

    for (int i = left; i <= right; i++) {

        int digit = arr[i] - '0';

        if (used[digit]) {
            continue;
        }

        used[digit] = true;

        char temp = arr[left];
        arr[left] = arr[i];
        arr[i] = temp;

        permute(arr, left + 1, right);

        temp = arr[left];
        arr[left] = arr[i];
        arr[i] = temp;
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    String input = sc.next();

    if (input.startsWith("-")) {
        input = input.substring(1);
    }

    char[] digits = input.toCharArray();

    permute(digits, 0, digits.length - 1);
}
```

Output

```
143
143
314
341
413
431
```